SHAZIN ANSARI

AI/ML Engineer | LLMs, RAG & Multi-Agent Systems | Python, FastAPI, LangChain, LangGraph | Bengaluru, India

shazinansari1@gmail.com | +91 9895236356 | LinkedIn | GitHub

Professional Summary

B.Tech (AI & Data Science) graduate and AI/ML engineer with 1+ year of hands-on experience building LLM and RAG applications in Python and evaluating frontier LLM and multimodal systems. Built RAG chatbots with LangChain, FAISS and ChromaDB, exposed through FastAPI REST APIs, containerized with Docker and backed by PostgreSQL and MySQL. Experienced in prompt engineering, LLM evaluation, hallucination reduction and faithfulness scoring, and in working with ML engineers and research teams to turn evaluation findings into model improvements. Comfortable with agentic tooling (LangGraph, LangSmith, Claude Code) and with documenting technical decisions clearly for technical and non-technical audiences.

Technical Skills

- **LLMs & Prompt Engineering:** LLM application development, prompt engineering, Hugging Face, LLaMA, Groq API, LLM evaluation, hallucination detection, faithfulness scoring
- **RAG & Agentic AI:** RAG architecture, LangChain, LangGraph, LlamaIndex, agentic AI and tool use, FAISS, ChromaDB, Pinecone, embeddings, chunking strategies, semantic search
- **Backend, APIs & SDK Integration:** Python, FastAPI, Flask, Django, RESTful API design, LLM provider SDK/API integration, microservices
- **Data & Pipelines:** Pandas, NumPy, EDA, feature engineering, data cleaning and preprocessing, document ingestion and chunking pipelines, MySQL, PostgreSQL
- **DevOps & Observability:** Docker, Git, CI/CD integration, LangSmith tracing, LLM observability
- **ML:** PyTorch, Scikit-learn, NLP, model evaluation (k-fold cross-validation, precision/recall)
- **AI Development Tools:** Claude Code, Antigravity, Lovable, AI-assisted development

Experience

Technical Research Associate – AI **Oct 2025 – Jun 2026**
Keywords Studios, Bengaluru

- Reduced hallucination rate and strengthened output safety, as measured by faithfulness scores fed into iterative training cycles, by partnering with ML engineers on structured LLM evaluation and benchmarking of AI-generated outputs.
- Improved instruction-following and factual accuracy, as measured by structured ranking scores across response-quality dimensions, by contributing to RLHF-style alignment workflows.
- Accelerated model refinement decisions, as measured by the number of systematic failure modes surfaced to research teams, by identifying, documenting and escalating edge cases in multimodal AI responses, including evaluation of agentic model behavior.
- Sustained consistent quality benchmarks across text and image modalities for multimodal foundation model training, as measured by team-wide output review coverage, by leading and coordinating task allocation for a team of 4 AI researchers.

AI Research Associate **Jul 2025 – Sep 2025**
Keywords Studios, Bengaluru

- Improved response quality and instruction adherence across state-of-the-art LLMs, as measured by consistency scores across diverse task types, by conducting structured prompt engineering and model evaluation.
- Shaped prompt design and model deployment decisions, as measured by evaluation insights adopted by cross-functional teams, by collaborating with GenAI teams on content generation and moderation workflows and communicating findings clearly in writing.
- Advanced RLHF and alignment research, as measured by volume of human preference feedback delivered, by evaluating LLM and multimodal AI outputs across text and image understanding tasks.

Data Science Intern **Jul 2024 – Apr 2025**
Future Optima IT Solutions, Kochi

- Achieved 85%+ classification accuracy on large datasets, as measured by model performance benchmarks, by cleaning data and building and optimizing decision tree and random forest models in Python with EDA and feature engineering.
- Maintained production model reliability, as measured by tracked accuracy and drift metrics, by collaborating with senior data scientists on algorithm implementation and ongoing model evaluation.

Projects

RAG-Based PDF Chatbot / *Python, FastAPI, LangChain, LangSmith, LLaMA, Groq API, FAISS, ChromaDB, Docker*

- Reduced response latency by 30% and improved answer faithfulness, as measured by LangSmith tracing and hallucination evaluation, by building a RAG application with a FastAPI REST backend, a document ingestion and chunking pipeline, a FAISS vector store, semantic search, and a Dockerized deployment for context-grounded responses.

Mental Health Chatbot / *Python, Flask, NLP, Semantic Search, REST API*

- Reached 85% query understanding accuracy across 50+ users, as measured by user testing, by designing dialogue flows and building a Python backend with RESTful APIs for an NLP chatbot using semantic intent matching.

Multi-Disease Prediction System / *Django, Scikit-learn, PyTorch, Python*

- Achieved 80% accuracy, 80% precision and 88% recall on multi-class disease prediction, as measured by k-fold cross-validation, by building the model pipeline and deploying a Django backend with CI/CD integration.

Education

B.Tech – Artificial Intelligence & Data Science **2020 – 2024**
MES College of Engineering, Kuttipuram | APJ Abdul Kalam Technological University

Certifications

- Generative AI with LangChain & Hugging Face – Krish Naik, Udemy (2025)
- Career Essentials in Generative AI – Microsoft & LinkedIn (2024)